12-22 第2篇

阅读理解（说明文） Why Does a Hot Drink Really Warm You on a Cold Day?

① On a cold December morning, after fifteen careful minutes of standing in the wind at a bus stop, you walk into a small corner café, wrap your hands around a paper cup of **hot ginger tea (about 70 °C)**, and take one slow sip. Within about **ninety seconds**, a small warm wave begins to move — not only through your fingers and mouth, but oddly, into your back, your shoulders, and even the tips of your ears. You feel, genuinely, warmer all over. Most of us accept this as obvious: *"I drank something hot, so of course I am warmer."* But the physics is, quietly, a small surprise. **The tea's thermal energy is actually very small; most of the warmth you feel does not come from the tea at all. It comes from your own body, gently switched on by the tea.**

② **Finding one: the tea contains much less heat than you think.** A standard 250 ml cup of tea at 70 °C, cooling down in your stomach to body temperature (37 °C), releases roughly **35 kilocalories (kcal)** of heat into your body. For reference, your body is already burning about **60—80 kcal per hour** simply by breathing and keeping your cells alive ("basal metabolic rate"). One hot cup of tea is therefore equal, in raw heat, to about **30 minutes of extra basal warmth** — spread out over the 30 minutes it takes for the tea to be absorbed and the water to cool. That is real, but it is tiny — nowhere near enough to warm your whole 50-kilogram body meaningfully by itself. And yet you feel warmer much faster than 30 minutes. Something else is happening.

③ **Finding two: the real warming is a trick of the skin.** When the hot liquid touches the **lining of your mouth and throat**, it activates a family of tiny heat sensors called **TRPV1 receptors** — the same receptors that detect the burn of a chili pepper. The sensors instantly send a signal to your brain's **hypothalamus**, the body's small "thermostat". The hypothalamus replies, in under a second, with a broad order throughout the body: *"External heat detected — open the blood vessels near the skin — bring warm blood to the surface — you are safe now."* Blood, which had been kept mostly in your deep core during the cold, suddenly flows outward in greater quantity to your hands, feet, face, and ears. This is called **vasodilation**. Your skin, previously cold and pale, fills with warm red blood. **You feel warmer — mainly because your own warm blood has come closer to your skin nerves.** The tea itself provided maybe 15% of the warmth; your own body provided about 85%.

④ **How big is this effect?** A **2018 University of Cambridge physiology study** measured skin surface temperature in **30 healthy young adults** after standing in a 5 °C cold room for 10 minutes and then drinking a 250 ml cup of either hot (70 °C) tea, lukewarm (40 °C) water, or icy (4 °C) water. After only **90 seconds**:

Drink	Skin temp. change (forearm)	Perceived warmth (1—10 scale)
Hot tea 70 ˚C	**+2.4 ˚C**	**+3.1**
Lukewarm water 40 ˚C	+0.2 ˚C	+0.4
Icy water 4 ˚C	**+1.1 ˚C**	**+0.6**

The surprising line is the last one. **Icy water also warmed the skin, by 1.1 ˚C on average.** Why? Because when a sudden cold signal arrived at the mouth, the hypothalamus briefly over-corrected — *"protect the core from sudden cold"* — and **temporarily increased blood flow to the skin as well**. The body, in other words, will warm your skin in response to many strong signals — including, sometimes, a cold one.

⑤ **Finding three: a hot drink actually cools your deep core very slightly.** This is the small strange fact that surprised researchers at the **University of Ottawa** in 2012. Their careful study of **9 male volunteers** on a stationary bicycle in a hot room (30 ˚C) showed that drinking a hot cup triggered **sweat** sooner than a cold one — and sweat evaporating from the skin cools the body **more than the hot tea itself heats it**. In a cold room, by contrast, the body almost never sweats, so this cooling effect is tiny and the warming effect of the vasodilation easily wins. That is why hot drinks feel warming only in cold environments. In a hot environment, in a perfectly strange way, a hot drink is slightly cooling.

⑥ **What about the "warm feeling" in my belly?** Part of that is real. The tea's direct 35 kcal **does** warm your stomach lining and the blood passing through it. The body's **vagus nerve** senses this gentle warming and sends a soft quieting signal — relaxing the muscles of the stomach and lowering heart rate slightly (a few beats per minute). This is the same feeling many of us describe with the slightly vague English word *cosy*. Cosiness, it turns out, is not a metaphor: it is your vagus nerve responding to a small warm wet signal in your stomach.

⑦ **Which hot drinks work best?** Any hot liquid gives the vasodilation response. But three classes of drinks add small extra effects:

Drink	Extra effect	Science note
Ginger tea	Activates TRPV1 more strongly; makes vasodilation larger and longer	Gingerol and 6-shogaol compounds bind to TRPV1; 2020 Kyoto University study
Black tea / coffee	Provides a mild attention lift from caffeine; helps with cold-morning sluggishness	Caffeine blocks adenosine receptors in the brain
Hot milk / cocoa	Provides small amount of real calories (about 150 kcal), sustaining body warmth over 60+ minutes	Protein and fat release energy slowly

Herbal tisanes without gingerol (plain chamomile, rose bud) give a smaller vasodilation boost but are still warming. **Plain hot water** works almost as well as tea, interestingly. It is the *hotness*, not the *tea-leaves*, that mainly warms the skin.

⑧ **Is it good for you to drink hot things on a cold day?** Mostly yes, with one small warning. **Chronic drinking of liquid above 65 °C** has been linked in several WHO reviews (2016, 2021) with a small increase in **oesophageal cancer risk** over decades. Most schools in China recommend drinking tea at **50—55 °C** — hot enough to feel deeply warming but not so hot that it damages the delicate cells lining the throat. The rule is simple: **a cup of tea should not make you wince when you sip.** If it does, wait two minutes.

⑨ **The quiet lesson.** Next time you come in from the cold and wrap your hands around a warm cup, remember: the cup is not doing most of the work. The cup is a small trigger. Your own body, quietly, is doing almost all of the warming — opening tens of thousands of tiny blood vessels near your skin, sending warm blood from your deep core out to your fingertips, your ears, your cheeks. **The tea is a small polite letter to your hypothalamus, and your hypothalamus, in about ninety seconds, writes a warm reply through your whole body.** Now you know. Drink the tea slowly. Your body is doing more for you than the tea is.

注释： TRPV1 瞬时受体电位 1 型；hypothalamus 下丘脑；vasodilation 血管舒张；vagus nerve 迷走神经；oesophageal 食管的；tisane 花草茶；gingerol / 6-shogaol 姜酚 / 姜烯酚

根据短文内容，选择最佳答案。

1. According to Paragraph ② and ③, why is the physics of "a hot drink warms you" a small surprise?

A. Because the tea contains much more heat than our body's basal metabolism.
B. Because a hot drink actually cools us instantly.
C. A 250 ml cup of tea at 70 °C cooling to body temperature (37 °C) releases only about 35 kilocalories — nowhere near enough by itself to warm a 50 kg body meaningfully, since the body is already burning 60—80 kcal per hour in basal metabolism; the real warming is a trick of the skin — the hot liquid touches the lining of the mouth and throat, activates TRPV1 heat sensors (the same that detect chili burn), which in under a second signal the hypothalamus (the body's thermostat); the hypothalamus orders vasodilation, so blood kept in the deep core during cold now flows outward in greater quantity to hands, feet, face, and ears, filling the skin with warm red blood — the tea provides maybe 15% of the warmth and your own body provides about 85%.
D. Because tea is a chemical that turns fingers red.

2. According to the 2018 University of Cambridge physiology study of 30 adults, what surprised the researchers and why?

A. Only hot tea warmed the skin; lukewarm water also warmed it strongly.
B. After 90 seconds of drinking, hot tea at 70 °C raised forearm skin temperature by about 2.4 °C and perceived warmth by 3.1 on a 1—10 scale; lukewarm water at 40 °C raised skin by only 0.2 °C; and surprisingly, icy water at 4 °C also warmed the skin by 1.1 °C, because when a sudden cold signal arrived at the mouth, the hypothalamus briefly over-corrected ("protect the core from sudden cold") and temporarily increased blood flow to the skin — so the body will warm your skin in response to many strong signals, including sometimes a cold one.
C. Icy water reduced skin temperature sharply.
D. Lukewarm water warmed the skin more than hot tea.

3. According to the 2012 University of Ottawa study, why can a hot drink actually be slightly cooling?

A. Because hot tea freezes the stomach.
B. Because in a hot room (30 °C), drinking a hot cup triggers sweat sooner than a cold drink, and evaporating sweat cools the body more than the hot tea itself heats it; in a cold room, the body almost never sweats, so this cooling is tiny and the vasodilation warming wins — which is why hot drinks feel warming only in cold environments, and in a hot environment, strangely, a hot drink is slightly cooling.
C. Because hot drinks contain ice crystals.
D. Because the tea turns into cold water inside the body.

4. What does Paragraph ⑦ say about different hot drinks and "plain hot water"?

A. Only ginger tea warms you; plain hot water does nothing.
B. All hot liquids give vasodilation, and three classes add extra effects: ginger tea activates TRPV1 more strongly (gingerol and 6-shogaol compounds binding to TRPV1, per 2020 Kyoto University study), making vasodilation larger and longer; black tea or coffee provides a mild attention lift from caffeine blocking adenosine receptors, helping with cold-morning sluggishness; hot milk or cocoa gives small real calories (about 150 kcal) that sustain body warmth over 60+ minutes, with protein and fat releasing energy slowly; herbal tisanes without gingerol (plain chamomile, rose bud) give a smaller vasodilation boost but are still warming; and plain hot water works almost as well as tea — it is the hotness, not the tea-leaves, that mainly warms the skin.
C. Cocoa never warms anyone.
D. Hot water is dangerous and should be avoided.

5. What is the main lesson of the final paragraph?

A. The cup of tea does all the warming.
B. Drinking tea is not healthy.
C. Hot drinks are a chemical trick by tea companies.
D. When you come in from the cold and wrap your hands around a warm cup, the cup is not doing most of the work but is a small trigger — your own body quietly does almost

all the warming, opening tens of thousands of tiny blood vessels near the skin and sending warm blood from the deep core out to the fingertips, ears and cheeks; the tea is a small polite letter to the hypothalamus, and the hypothalamus in about ninety seconds writes a warm reply through the whole body — so drink the tea slowly, because the body is doing more for you than the tea is.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②③段茶热量 + 皮肤血管舒张
【解析】"35 kcal / TRPV1 / 下丘脑 / 15% vs 85%"直引。A、B、D 均错。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第④段剑桥 2018 表格
【解析】"70 ℃ +2.4 / 4 ℃ +1.1 / 冷也触发血流"直引。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段渥太华 2012 研究
【解析】"热汽房诱汗 / 蒸发降温 / 冷房不出汗 / 热室反致凉"直引。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑦段饮品比较
【解析】"姜茶 TRPV1 / 咖啡因 / 奶 150 kcal / 白水也行 / 关键是热"直引。A、C、D 均错。故选 B。

5.
【答案】D
【题型】主旨大意题
【定位】第⑨段结尾
【解析】"杯是触发器 / 身体做大部分 / 礼貌信件 / 慢慢喝"主旨。A、B、C 均偏离。故选 D。